

Thermal Transport in Cylindrical Channels

Non-Uniform Wall Boundary Conditions

and Large Aspect Ratio Analysis

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Introduction

These notes develop a perturbative analysis of how thermal non-uniformity affects the flow profile in laminar cylindrical channels, motivated by transformer cooling applications.

The goal is to address a gap in the electrification literature. Analyses of charge transport in transformer oil typically assume isothermal flow with constant physical properties (viscosity, thermal and electrical conductivity, density). We relax this assumption and study the leading-order effects of temperature-dependent properties on both flow and later transport.

We acknowledge the limitations: real transformer temperature gradients are often too large for perturbation theory to be quantitatively accurate. This work is not meant to replace CFD simulation, but rather to provide an analytical framework for understanding first-order coupling mechanisms and guiding numerical predictions.

To achieve this goal, we started by solving the isothermal flow profile. This result is then used in the energy equation to derive the temperature distribution in the fluid under non homogeneous boundary conditions. These results are coupled back to the momentum equations through the fluid properties dependence on temperature allowing for a computation of the first order correction to the velocity fluid.

Scope. These notes cover:

- momentum equation derivation in cylindrical coordinates
- Energy equation derivation in cylindrical coordinates
- Asymptotic analysis using the aspect ratio
- Particular solution (viscous dissipation)
- Homogeneous solution (boundary condition response)
- Fourier series expansion and boundary matching

- Test cases (uniform and non-uniform BC)
- Physical interpretation and connection to charge transport

Notation. Starred quantities (T^* , r^* , z^* , etc.) denote dimensionless variables. The asterisk is sometimes suppressed for brevity. The Laplacian operator in cylindrical coordinates appears as ∇^2 (full 3D) or $\nabla_{\perp}^2$ (radial + angular only).

Contents

Introduction	I
1 Isothermal Velocity Field	4
1.1 Physical Setup & Symmetry Considerations	4
1.2 Momentum transport & Velocity profile	4
2 Energy Balance & Temperature Profile	6
2.1 Energy Transport Mechanisms & the First Law of thermodynamics	6
2.2 Dimensionless equation & Reference Scales	9
2.3 Small Aspect Ratio limit & Temperature Profile	10
3 Test Cases and Validation	II
3.1 Case 1: Uniform Wall Temperature	II
3.2 Case 2: One-Sided Cooling (Non-Uniform)	II
4 Physical Interpretation	12
4.1 Asymptotic Structure and Scaling	12
4.2 Relative Importance of Modes	12
4.3 Penetration Depth of Non-Uniformity	12
4.4 Connection to Charge Transport	12
5 Order-ϵ Corrections and Next Steps	12
5.1 What is Captured at Order 0	12
5.2 What Requires Order- ϵ Corrections	13
5.3 Implementation and Validation	13
6 Summary and Outlook	13
6.1 What Was Solved	13
6.2 Limitations	13
6.3 Path Forward	13
A Cylindrical Laplacian	14

I Isothermal Velocity Field

I.1 Physical Setup & Symmetry Considerations

Consider a long cylindrical cooling channel of radius R and length L , filled with a viscous fluid (e.g., transformer oil). The fluid enters at a uniform inlet temperature and flows under a pressure gradient Δp . Table I summarizes the different physical properties of the fluid and the symbols used to represent them.

Physical property	symbol
Density	ρ
viscosity	μ
Thermal Capacity per unit mass at constant pressure	$\hat{C}_p$
Thermal conductivity	K
Thermal diffusivity	α

Table I: Summary of the fluid physical properties and symbols

Assuming that the length of the cooling channel L is much larger than the radius R , we expect that the velocity field to be predominantly axial along the z direction with only radial dependence. Hence for the isothermal case we need only consider $v_z(r)$.

I.2 Momentum transport & Velocity profile

We first start by defining the relevant mathematical objects for the derivation of the momentum transport equations. There are three different mechanism responsible for momentum transport for a flowing system

- First, transport through the bulk movement of the fluid. This is represented formally through *The Convective Momentum Transport tensor* $\phi_{ij} = \rho v_i v_j$.
- Second, momentum transport through molecular motion. This is taken into account through *The Shear Stress tensor*, τ_{ij} defined by

$$\tau_{ij} = -\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \left(\frac{2}{3}\mu - \kappa \right) \nabla \cdot \mathbf{v} \delta_{ij}. \quad (1)$$

κ is the bulk viscosity of the fluid. For this analysis, we will only consider incompressible fluids verifying the equation $\nabla \cdot \mathbf{v} = 0$. Hence the second term in the previous expression vanishes.

- Finally, momentum can be transported through pressure gradients.

To take into account all three forms of momentum transport we define the *Combined Momentum Tensor*, Φ_{ij}

$$\Phi_{ij} = \rho v_i v_j + \tau_{ij} - P \delta_{ij}. \quad (2)$$

Physical Remark. The tensor element Φ_{ij} can be interpreted as the transport rate of the j -th component of momentum across a surface orthogonal to the i direction.

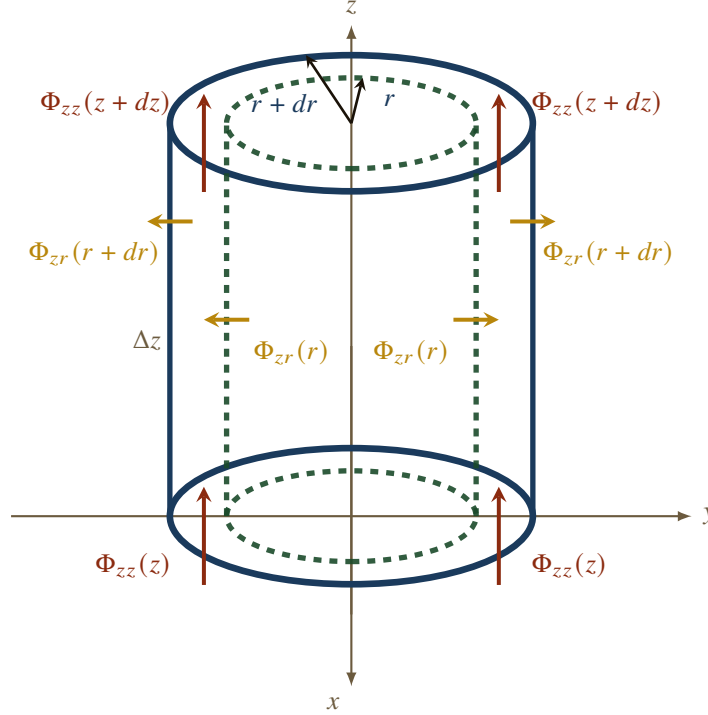


Figure 1: Cylindrical shell element showing the annular region between radius r (inner, dashed) and radius R (outer, solid) with axial extent Δz . The combined momentum tensor flux is shown.

The shell balance method consists of equating the rate of momentum change for a volume shell to the flux of the tensor through its boundaries. Given that we consider a steady state flow, we thus impose a null momentum balance through the shell boundaries.

Derivation 1.1 (Momentum Transport Equation). Consider the volume shell represented in figure 1. Given the symmetries of the system and the postulated velocity field form the only non vanishing terms of Φ are Φ_{zz} and ϕ_{zr} . The momentum balance for the z component is given by

$$2\pi r dr (\Phi_{zz}(z) - \Phi_{zz}(z + dz)) + (2\pi r dz \Phi_{rz}(r) - 2\pi(r + dr) dz \Phi_{rz}(r + dr)) = 0 \quad (3)$$

Dividing by the volume element $dv = 2\pi r dr dz$ and taking the limit $dr \rightarrow 0$, $dz \rightarrow 0$ leads to the following differential equation

$$-\frac{\partial \Phi_{zz}}{\partial z} - \frac{\partial \Phi_{rz}}{\partial r} - \frac{1}{r} \Phi_{rz} = 0. \quad (4)$$

Expressing the dependence of Φ explicitly on the velocity field,

$$\Phi_{zz} = \rho v_z(r)^2 - P(z), \quad (5a)$$

$$\Phi_{rz} = \tau_{rz} = -\mu \partial_r v_z, \quad (5b)$$

we get the momentum transport equation verified by the velocity field,

$$\boxed{+\partial_z P = -\mu \frac{1}{r} \partial_r (r \partial_r v_z)} \quad (6)$$

Integrating this equation leads to

$$v_z(r) = -\frac{\Delta P}{4\mu} r^2 + c \ln(r) + c'. \quad (7)$$

c and c' are integration constant. Their value is determined through the no slip boundary condition at $r = R$, $v_z(R) = 0$, and requiring the centerline velocity $v_z(0)$ is finite. Hence we get

$$\boxed{v_z(r) = v_0 \left(1 - \left(\frac{r}{R}\right)^2\right)}, \quad (8)$$

with $v_0 = \frac{\Delta P}{4\mu} R^2$.

Physical Remark (remarks on Poiseuille Velocity Profile). Equation 8, is the standard Poiseuille flow result. The velocity field is parabolic with a maximum velocity at the center of the channel. The shear stress is maximum at the boundary, $r = R$, and equal to $\tau_{|wall} = \frac{2\mu v_0}{R}$.

Regarding the derivation of the equation, one should note that, in general, the continuity equation ($\nabla \cdot \mathbf{v} = 0$ for incompressible fluids) is also needed. The shell balance techniques for momentum transport provides three equations, one for each on the momentum components. The total number of unknowns on the other hand is four, the three velocity components and pressure. In the derivation of the velocity field the pressure gradient along the z -direction was not specified nor solved for, this is because we assume an experimental setup where the pressure gradient is controlled.

2 Energy Balance & Temperature Profile

With the isothermal steady state velocity at hand, we now consider the case where the boundary of the channel, $r = R$, has a non-uniform temperature distribution both in the azimuthal and the longitudinal direction, $T(R, \theta, z) = T_{BC}(\theta, z)$. The non-uniform boundary condition break the axial symmetry.

2.1 Energy Transport Mechanisms & the First Law of thermodynamics

In this analysis we neglect temperature transport through radiation and focus mainly the the diffuse and convective mechanisms. Thus, let $\mathbf{q}$ be the *molecular diffusion energy flux*, it represents the energy

flux due to the thermal movement of molecules. It is related to the temperature field through Fourier's law, $\mathbf{q} = -K \nabla T$, with K the thermal conductivity of the fluid.

To derive the convective energy transport, consider a fluid volume dV moving with a velocity $\mathbf{v}$. The energy contained in this volume is given by

$$E = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) dV, \quad (9)$$

where $\hat{U}$ is the internal energy per unit volume of the fluid. the energy flux across a surface element $d\mathbf{s}$ orthogonal to the x direction is then

$$Q = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) v_x dV ds, \quad (10a)$$

$$= \mathbf{c} \cdot d\mathbf{s} \quad (10b)$$

Where $\mathbf{c}$ is the *convective energy flux*, $\mathbf{c} = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) \mathbf{v}$.

Finally, the derivation of the energy equation requires the application of the first law of thermodynamics for an open system. This is achieved by requiring that the total energy variation for a volume element is equal to the heat intake and to the work of the molecular forces. Consider a cube of volume dV and let ds be one of its sides at a constant x value, the force applied at this surface is $\Pi_x ds$ with $\Pi_x)_i = \tau_{xi} + P\delta_{xi}$. Where τ_x represents the shear forces applied at the surface and P the pressure. Thus the rate of the work applied at the surface is given by $\frac{dw}{dt} = \Pi_x \cdot \mathbf{v} ds$. Hence for a general surface give by the normal $d\mathbf{s}$. The work rate is $(\Pi \cdot \mathbf{v}) \cdot d\mathbf{s}$.

Remark. The operation $\cdot$ here is used in generalized form as a dot product between tensors of different ranks. Thus in the expression of the work rate it take a second rank tensor Π and a first rank tensor $\mathbf{v}$ and produces a first rank tensor.

In a similar manner to the *combined momentum transport tensor* Φ , we define the *combined energy flux vector* $\mathbf{e}$ by

$$\mathbf{e} = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) \mathbf{v} + \Pi \cdot \mathbf{v} + \mathbf{q}. \quad (11)$$

The flux of this vector field along a closed surface represents to the total energy variation of the surface. We shall express $\mathbf{e}$ in a more useful form using the enthalpy $\hat{H} = \hat{U} + P \hat{V}$, where $\hat{V} = \frac{1}{\rho}$ is the volume per unit mass of the fluid. Expanding Π in pressure and shear contribution, we get

$$\mathbf{e} = \left(\frac{1}{2} \rho v^2 + \rho \left(\hat{U} + P \hat{V} \right) \right) \mathbf{v} + \tau \cdot \mathbf{v} + \mathbf{q}, \quad (12a)$$

$$= \left(\frac{1}{2} \rho v^2 + \rho \hat{H} \right) \mathbf{v} + \tau \cdot \mathbf{v} + \mathbf{q} \quad (12b)$$

Derivation 2.1 (Temperature transport equation). Given the dependence of the boundary conditions on the azimuthal angle θ , we consider the shell shown in figure 2. The energy balance

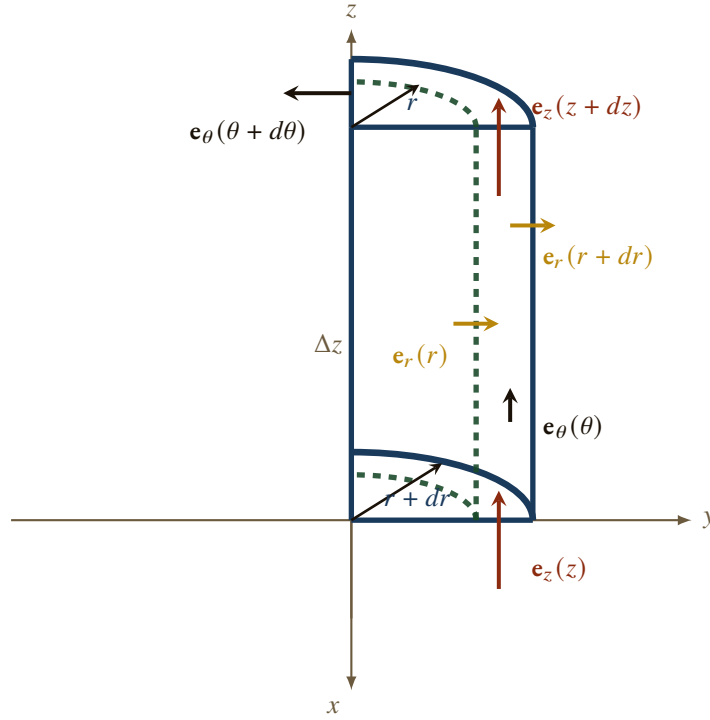


Figure 2: Combined energy flux through a cylindrical volume element.

equation is given by

$$r \, dr \, d\theta \, (e_z(z) - e_z(z + dz)) + r \, dz \, d\theta \, (e_r(r) - e_r(r + dr)) - dr \, d\theta \, dz \, e_r(r + dr) \quad (13a)$$

$$+ dr \, dz \, (e_\theta(\theta) - e_\theta(\theta + d\theta)) = 0. \quad (13b)$$

Dividing by the volume element $dV = r \, dr \, d\theta \, dz$ leads to

$$-\partial_z e_z - \frac{1}{r} \partial_r (r \partial_r e_r) - \frac{1}{r} \partial_\theta e_\theta = 0 \quad (14)$$

Using the velocity profile 8, we can combine each derivative appearing in the energy balance equation.

- Given $e_z = \left(\frac{1}{2} \rho \mathbf{v}^2 + \rho \hat{H} \right) v_z + q_z$, we get

$$\partial_z e_z = \rho \partial_z \hat{H} + \partial_z q_z. \quad (15)$$

The dependence of the enthalpy on coordinates is implicit through its dependence on temperature and pressure. Thus,

$$\partial_z \hat{H} = \partial_z T \, \partial_T \hat{H} + \partial_z P \, \partial_P \hat{H}. \quad (16)$$

Using the following thermodynamics identities,

$$\partial_T \hat{H} = \hat{c}_p \quad , \quad \partial_P \hat{H} = \hat{V} + \rho \partial_P \hat{V}, \quad (17)$$

and assuming a constant fluid density, we get

$$\partial_z \hat{H} = \hat{c}_p \partial_z T + \partial_z P. \quad (18)$$

Fourier's law indicates that $q_z = -K \partial_z T$. Putting everything together we get

$$\partial_z e_z = (\rho \hat{c}_p \partial_z T + \partial_z P) v_z - K \partial_z^2 T \quad (19)$$

- For the radial component of the or the combined energy flux vector, we have

$$e_r = \tau_{rz} v_z + q_r \quad (20)$$

Thus

$$\frac{1}{r} \partial_r (r \partial_r e_r) = -\frac{\mu}{2} \frac{1}{r} \partial_r (r \partial_r v_z^2) - \frac{K}{r} \partial_r (r \partial_r T) \quad (21)$$

- Finally, $e_\theta = q_\theta$, hence

$$\frac{1}{r} \partial_\theta e_\theta = -\frac{K}{r^2} \partial_\theta^2 T \quad (22)$$

The energy balance equation takes the form

$$K \nabla^2 T = \rho \hat{c}_p v_z \partial_z T - v_z \partial_z P + \frac{\mu}{2} \frac{1}{r} \partial_r (r \partial_r v_z^2). \quad (23)$$

This equation can be further simplified using the momentum transport equation 6.

$$-v_z \partial_z P + \frac{\mu}{2} \frac{1}{r} \partial_r (r \partial_r v_z^2) = -v_z \underbrace{\left[\partial_z P - \mu \frac{1}{r} \partial_r (r \partial_r^2 v_z) \right]}_{=0} + \mu (\partial_r v_z)^2 \quad (24)$$

Finally the heat equation takes the following form

$$\boxed{\rho \hat{c}_p v_z \partial_z T = K \nabla^2 T - \mu (\partial_r v_z)^2} \quad (25)$$

2.2 Dimensionless equation & Reference Scales

Before solving the heat transport equation 25, we first express it in a dimensionless form. To this end define the dimensionless parameters 2. Here we have defined the kinetic temperature $T_k = \frac{v_o^2}{2\hat{c}_p}$. In this new parameters the temperature transport equation takes the form

$$v^* \partial_{z^*} T^* = \frac{\alpha}{Lv_o} \left[\partial_{z^*}^2 T^* + \left(\frac{L}{R} \right)^2 \frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \left(\frac{L}{R} \right)^2 \frac{1}{r^{*2}} \partial_{\theta^*}^2 T^* \right] - 8 \frac{\mu}{\rho Lv_o} \left(\frac{L}{R} \right)^2 r^{*2} \quad (26)$$

Variable	Expression
r^*	$\frac{r}{R}$
v^*	$\frac{v_z}{v_o}$
z^*	$\frac{z}{L}$
T^*	$\frac{T}{T_k}$

Table 2: Dimensionless parameter definition

The parameter $\alpha = \frac{K}{\rho \hat{c}_p}$ is the temperature diffusivity of the fluid. Finally using the definition of the Péclet, Reynolds, Prandtl numbers and the aspect ratio,

$$\text{Pe} = \frac{Lv_o}{\alpha}, \quad \text{Re} = \frac{\rho R v_o}{\mu}, \quad \epsilon = \frac{R}{L}, \quad \text{Pr} = \frac{\text{Pe}}{\epsilon \text{Re}} \quad (27)$$

we get the following equation

$$v^* \epsilon^2 \partial_{z^*} T^* = \frac{1}{\text{Pe}} \left[\epsilon^2 \partial_{z^*}^2 T^* + \frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \frac{1}{r^{*2}} \partial_{\theta}^2 T^* \right] - \frac{8}{\text{Re}} \epsilon r^{*2}. \quad (28)$$

2.3 Small Aspect Ratio limit & Temperature Profile

Equation 28 indicates a clear scale separation between the azimuthal and radial variations of the temperature field and the longitudinal ones. Thus considering the case of small aspect ratio $\epsilon \rightarrow 0$ we can ignore the longitudinal variation of the temperature the equation takes then the simplified form.

$$\frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \frac{1}{r^{*2}} \partial_{\theta}^2 T^* = 8 \text{Pr} r^{*2}. \quad (29)$$

Derivation 2.2 (Solving the Temperature Transport Equation). To solve the previous equation we first consider the homogeneous equation without the shear source

$$\frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \frac{1}{r^{*2}} \partial_{\theta}^2 T^* = 0. \quad (30)$$

Separating the variables,

$$T^*(r^*, z^*, \theta) = f(r^*, z^*) g(\theta), \quad (31)$$

equation ?? separates into two equations

$$\frac{d^2 g}{d\theta^2} + \omega^2 g(\theta) = 0 \quad (32a)$$

$$\frac{1}{r^*} \partial_{r^*} (r \partial_{r^*} f) = \omega^2 f. \quad (32b)$$

The first equation leads to $g(\theta) = e^{i\omega\theta}$ then imposition the periodicity condition $g(\theta) = g(\theta + 2\pi)$ leads to $\omega = m \in \mathbb{Z}$. The second equation is solved by expressing f as power series $f(r^*, z^*) = \sum_{n \in \mathbb{Z}} a_n(z^*) (r^*)^n$, the second equation then imposes that $n^2 = m^2$. However the

demand that the temperature at $r^* = 0$ stay finite leads to $n = |m|$. Thus, a general solution for the homogeneous equation is

$$T^*(r^*, z^*, \theta) = \sum_{m \in \mathbb{Z}} a_m(z) r^{*|m|} e^{im\theta}. \quad (33)$$

To obtain a solution of the temperature transport equation we need to find a particular solution to 29. Thus, let $T_p(r^*)$ be a solution. We get

$$T_p(r^*) = \frac{\text{Pr}}{2} r^{*4} \quad (34)$$

Hence the temperature field is

$$T^*(r^*, z^*, \theta) = \frac{\text{Pe}}{2\text{Re}} r^{*4} + \sum_{m \in \mathbb{Z}} a_m(z) r^{*|m|} e^{im\theta}. \quad (35)$$

The coefficients $a_m(z)$ are determined through the boundary conditions by

$$a_m(z) = \begin{cases} a_0(z) = \frac{1}{2\pi} \int_0^{2\pi} T_{BC}(\theta, z) d\theta - \frac{\text{Pr}}{2} \\ a_m(z) = \frac{1}{2\pi} \int_0^{2\pi} T_{BC}(\theta, z) e^{-im\theta} d\theta, m \neq 0 \end{cases} \quad (36)$$

3 Test Cases and Validation

3.1 Case 1: Uniform Wall Temperature

Example 3.1 (Baseline Case). Suppose the wall is isothermal: $T_w(\theta) = T_o$ (constant).

Then:

$$a_m = \frac{1}{2\pi} \int_0^{2\pi} (T_o - 1/4) e^{-im\theta} d\theta = \begin{cases} T_o - 1/4 & m = 0 \\ 0 & m \neq 0 \end{cases}$$

The solution becomes:

$$T^* = \frac{r^{*4}}{4} + (T_o - 1/4).$$

This is purely axisymmetric and matches the classical result for Poiseuille flow with uniform wall temperature.

3.2 Case 2: One-Sided Cooling (Non-Uniform)

Example 3.2 (Realistic Non-Uniform BC). Suppose cooling is stronger on one side:

$$T_w(\theta) = T_o + \Delta T \cos(\theta).$$

Then:

$$a_m = \frac{1}{2\pi} \int_0^{2\pi} [\Delta T \cos \theta] e^{-im\theta} d\theta.$$

By the orthogonality of Fourier modes:

$$a_m = \begin{cases} T_o - 1/4 & m = 0 \\ \Delta T/2 & m = \pm 1 \\ 0 & |m| \geq 2 \end{cases}$$

The solution is:

$$T^* = \frac{r^{*4}}{4} + (T_o - 1/4) + \frac{\Delta T}{2} (r^* e^{i\theta} + r^* e^{-i\theta}) = \frac{r^{*4}}{4} + (T_o - 1/4) + \Delta T r^* \cos \theta.$$

Physical interpretation:

- The $r^{*4}/4$ term is the dissipation baseline.
- The constant $(T_o - 1/4)$ is the uniform cooling response.
- The $r^* \cos \theta$ term shows the asymmetry penetrates radially with amplitude proportional to r^* — it decays toward the axis.
- Higher harmonics ($m \geq 2$) are automatically zero, reflecting that the BC only has one-sided structure.

4 Physical Interpretation

4.1 Asymptotic Structure and Scaling

4.2 Relative Importance of Modes

4.3 Penetration Depth of Non-Uniformity

4.4 Connection to Charge Transport

5 Order- ϵ Corrections and Next Steps

5.1 What is Captured at Order 0

Remark. The Order-0 solution $T_o^*(r^*, \theta)$ is independent of z^* and fully captures the cross-sectional temperature structure. The viscous dissipation is included at leading order.

Discuss why $\epsilon = R/L$ naturally separates scales

Compare relative importance of dissipation, radial/angular diffusion

Discuss the role of different Fourier modes

For your oil parameters, compute a_0 , a_1 , a_2 , etc.

Show which

5.2 What Requires Order- ϵ Corrections

Remark.

- The particular solution $T_p(r^*)$ may acquire a z^* dependence at Order ϵ due to the pressure-work term and finite Péclet effects.
- The coefficients $a_m(z^*)$ will evolve slowly if the wall BC varies with z .
- Axial conduction becomes relevant at very long distances.

The next stage would set up the Order- ϵ equations. For now, the leading-order solution is complete and can be used directly.

5.3 Implementation and Validation

6 Summary and Outlook

6.1 What Was Solved

6.2 Limitations

6.3 Path Forward

Write pseudocode for evaluating $T^*(r^*, \theta)$ numerically

Outline comparison with FEM solver (COMSOL)

Discuss convergence: how many modes m are needed?

Bullet-point summary of the solution

Order ϵ corrections not included

Velocity profile still decoupled (not coupled back to temperature)

Fluid properties assumed constant

Stage 2: Thermal

References

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A Cylindrical Laplacian

The full Laplacian in cylindrical coordinates is:

$$\nabla^2 T = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2}.$$

The transverse (radial + angular) Laplacian is:

$$\nabla_{\perp}^2 T = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2}.$$